

# Debayan Biswas

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IIT Kharagpur, Physics. Backend and systems engineer: production analytics platforms, Java agentic services, vector search and data pipelines, LLM and MCP integrations.

## EDUCATION

### Indian Institute of Technology, Kharagpur

Kharagpur, West Bengal

*Bachelors Degree, Physics*

*Apr. 2025*

## EXPERIENCE

### Software Engineer

May 2025 – Present

*Cimba.ai*

*Kolkata, West Bengal*

- Led Usage Insights end to end: unified POST dispatcher, KPI and trend APIs (latency, tokens, failures, users, tasks), server-side pagination, presigned CSV export, React and ECharts in production, consolidated analytics response models.
- Decommissioned Prediction Service and Pinecone; shipped Java agentic runtime, Azure AI Search (**migrateAgent**), and async knowledge-base indexing with live status in production.
- Built WebSocket flows for Deep Research (memory, notifications, completion) and RequestId propagation from test runner through API to FDS for cross-service tracing.
- Shipped MCP OAuth (entity-scoped tokens, sharing, refresh), Redis caching on chat and API key validation, and Google Sheets to DuckDB ingestion.
- Re-architected Ask Cimba tests from a ~700-line Python runner to modular Java; cut duplicate YAML 71% by consolidating 55 provider files into 11 question sets and 5 configs.
- Owned CI/CD for LLM regression with three-channel Slack across five providers and isolated follow-up cases from general traffic.
- Shipped Enrich (JSON, CSV, XLSX, async migration, related-table enrichment), server-side agent-builder tables, and notification preferences that retired the legacy email bypass.

### Software Engineer Intern

Jun. 2024 – Jul. 2024

*Ittiam Systems*

*Bangalore, Karnataka*

- Fixed A/V sync in a production video pipeline using FFmpeg split-and-stitch over chunked media.
- Built a Minikube Kubernetes dev stack (Ingress, metrics, registry) and Docker images for repeatable integration tests.
- Ran a distributed encode pipeline: NFS-backed storage, KEDA-driven autoscaling, Redis and BullMQ workers sized on queue depth.

### Software Engineer Intern

Dec. 2023 – Apr. 2024

*OneLot*

*Manila, Philippines*

- Redesigned parts of the product UI, built forms for key flows, segregated Next.js server versus client components, and added custom React hooks to reuse logic and optimize client rendering and state.
- Integrated Slack into the website for team communication (15% collaboration lift) and contributed to SEO initiatives associated with 25% more SEO-attributed traffic.

## PROJECTS

### IntentFlow | *Go, PostgreSQL, OpenAPI, MCP, LLM APIs*

Feb. 2026

- Designed IntentFlow control plane in Go: OpenAPI-backed intent, plan, confirm, and deterministic local execution on Postgres with JWT and RBAC (compiler-style workflow, not an open agent loop).
- Implemented a DAG executor (parallel steps, **depends\_on**), registry LLM backends (OpenAI, Gemini, Hugging Face Inference), and MCP orchestration in TUI/API with per-user isolation and token refresh.
- Added CI pipelines, golanci-lint, Dockerized Postgres, and teatest end-to-end coverage (~73 commits).

## TECHNICAL SKILLS

**Backend & systems:** Java, Go, OpenAPI, REST, distributed systems, event-driven workers, WebSockets

**Data & search:** PostgreSQL, Redis, DuckDB, Azure AI Search, NFS

**Infra & delivery:** Docker, Linux, Kubernetes (Minikube), KEDA, BullMQ, FFmpeg, CI/CD, DataDog

**Languages:** Scala, Python, JavaScript, TypeScript, C/C++

**Frontend:** React, Next.js, Node, ECharts

**Security & AI:** JWT, RBAC, MCP, LLM APIs